

TRIGONOMETRY REVISION

[ANSWERS]

(1) a) 17.3 cm b) 10 cm c) 7.3 cm d) 24.5 cm

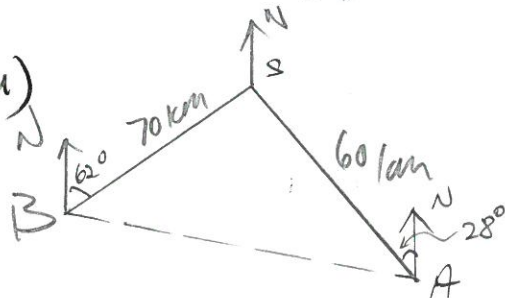
(2) 37.1 cm (3) 8.2 cm

(4) $\theta = 42.0^\circ$, $x = 7.8$ cm (5) 5.71 (2 d.p.)

(6) a) 26.7 cm b) 89.4° (7) 29.28 m

(8) 68.67 m

(9) a)



b) 92.2 km

c) 077.4°

(10) 568.34 m (2 d.p.) (11) 27.96 cm (12) 34.85°

(13) a) 9.2 m (1 d.p.) b) 3.6 m (1 d.p.) (14) 29.24 cm

(15) $\angle BAC = 65^\circ$, $AC = 39.1$ mm, $BC = 36.7$ mm

(16) $\angle CBA = 59^\circ$, $AC = 34.4$ cm, $BC = 40.0$ cm

(17) $\angle BAC = 65^\circ$, $\angle ACB = 35^\circ$, $AB = 28.5$ cm

(18) $\theta = 69.5^\circ$ (1 d.p.) (19) $\angle BAC = 65 - 90^\circ$, $\angle ABC = 74.2^\circ$
 $\angle ACB = 39.9^\circ$

(20) a) 25.5 m b) 52° c) 318.2 m² d) 608 m²

(21) 8.04 m (22) 9.25 m

(23) $ADC = 24.32$ km $ADB = 26.11$ km

ADC is a better option because its shorter distance implies lower cost.

(24). $\angle CAB = 85.0^\circ$ $\angle ABC = 59.0^\circ$ $\angle BCA = 36.0^\circ$

(25). 18 m (26). a) 122.5° b) 22.05 m

(27) a) 60.8 m b) 40° c) 1003.6 m^2 d) 3362 cm^2

(28) $\angle ABC = 44.0^\circ$ $\angle BCA = 64.0^\circ$ $AB = 43.5 \text{ mm}$

(29) $\angle CAB = 41.8^\circ$ $\angle BCA = 78.2^\circ$ $c = 14.7 \text{ cm}$

(30) $\angle QRP = 82^\circ$ $PQ = 205.0 \text{ cm}$ $QR = 80.9 \text{ m}$

(31) $\angle BAC = 67.0^\circ$ $\angle CBA = 48.0^\circ$ $\angle ACB = 65.0^\circ$

(32) $\angle BAC = 80.4^\circ$ $\angle CBA = 45.8^\circ$ $\angle ACB = 53.8^\circ$